

Lab 2: Wireless Network Security – Wireshark Lab Workbook

INFT 1101 – Fuad Mustapha

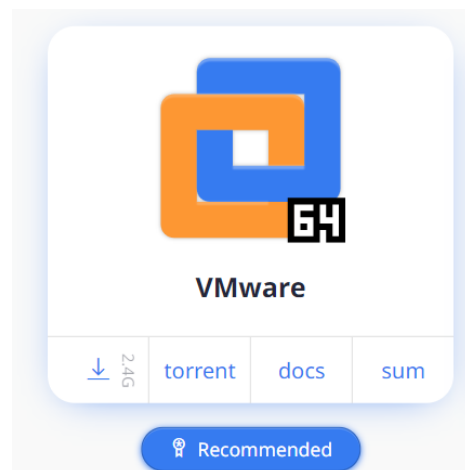
Madhvi Lashkaree (100997946)

Introduction

Wireshark, a packet sniffer, is introduced in this second Lab. Since you have your kali Linux, you would have Wireshark installed already. Wireshark is an open-source network protocol analyzer that is free to use. It is used to troubleshoot networks and analyze communication protocols. Wireshark captures network packets in real time and displays them in a human-readable format. It has many advanced features, such as live capture and offline analysis, three-pane packet browser, coloring of rule analysis. This document describes Wireshark installation and usage for the Lab, Protocol analysis and packet capture.

Pre-requisites of the lab

1. VMware
2. RAM 2+GB
 - i. 2 GHz dual core processor or better
 - ii. 4 GB system memory
3. 20 GB of free hard drive space
4. Internet access is mandatory to access Durham colleges website
5. Kali VMware ova image (the easy way to setup kali as a VM) can be obtained here: <https://www.kali.org/get-kali/#kali-virtual-machines>

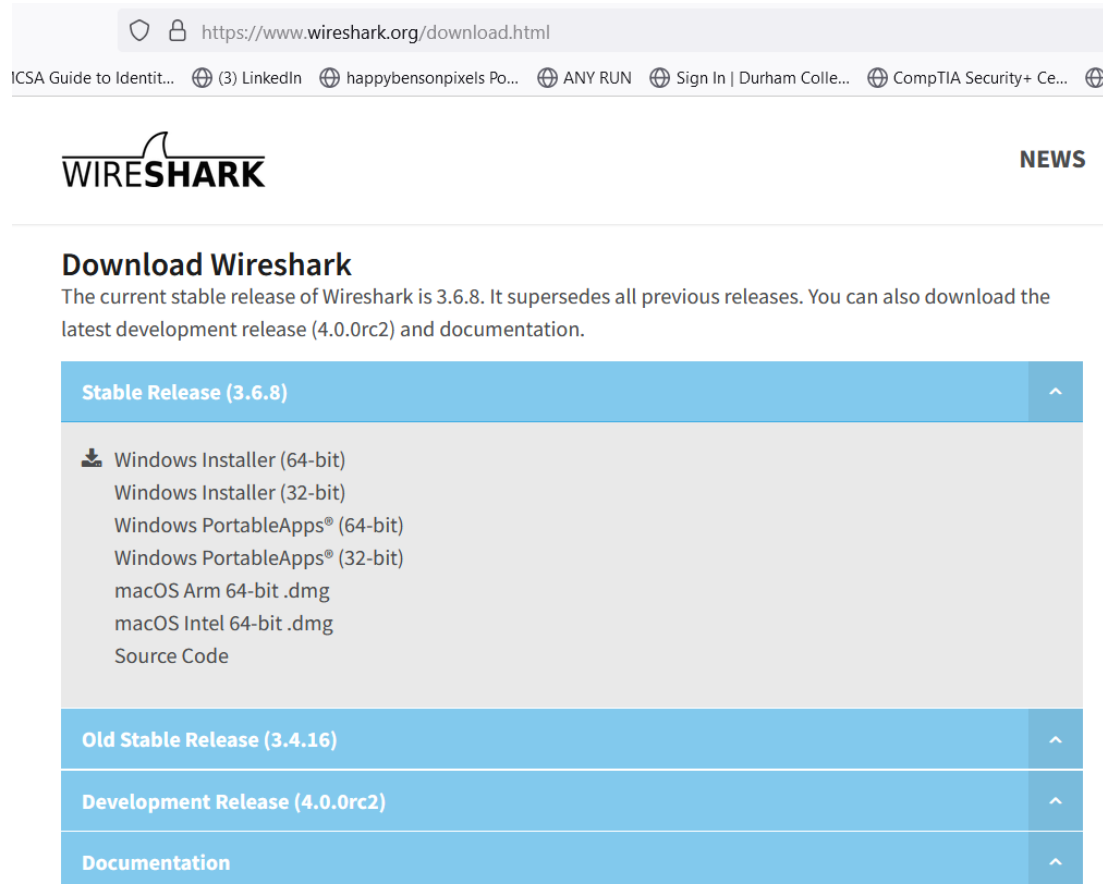


Lab 2: Packet Sniffing and Wireshark

Getting Wireshark

The Kali Linux has Wireshark installed. You can just launch the Kali Linux VM and open Wireshark there. Wireshark can also be downloaded from here:

<https://www.wireshark.org/download.html>



The screenshot shows the Wireshark website's download page. At the top, there's a navigation bar with the Wireshark logo on the left and the word "NEWS" on the right. Below the navigation bar, the heading "Download Wireshark" is followed by a paragraph stating that the current stable release is 3.6.8, which supersedes all previous releases, and that users can also download the latest development release (4.0.0rc2) and documentation. Below this text is a list of download options, each in a blue box with a right-pointing arrow. The first box, "Stable Release (3.6.8)", is expanded to show a list of download links: Windows Installer (64-bit), Windows Installer (32-bit), Windows PortableApps® (64-bit), Windows PortableApps® (32-bit), macOS Arm 64-bit .dmg, macOS Intel 64-bit .dmg, and Source Code. The other boxes, "Old Stable Release (3.4.16)", "Development Release (4.0.0rc2)", and "Documentation", are collapsed.

Stable Release (3.6.8)

- Windows Installer (64-bit)
- Windows Installer (32-bit)
- Windows PortableApps® (64-bit)
- Windows PortableApps® (32-bit)
- macOS Arm 64-bit .dmg
- macOS Intel 64-bit .dmg
- Source Code

Old Stable Release (3.4.16)

Development Release (4.0.0rc2)

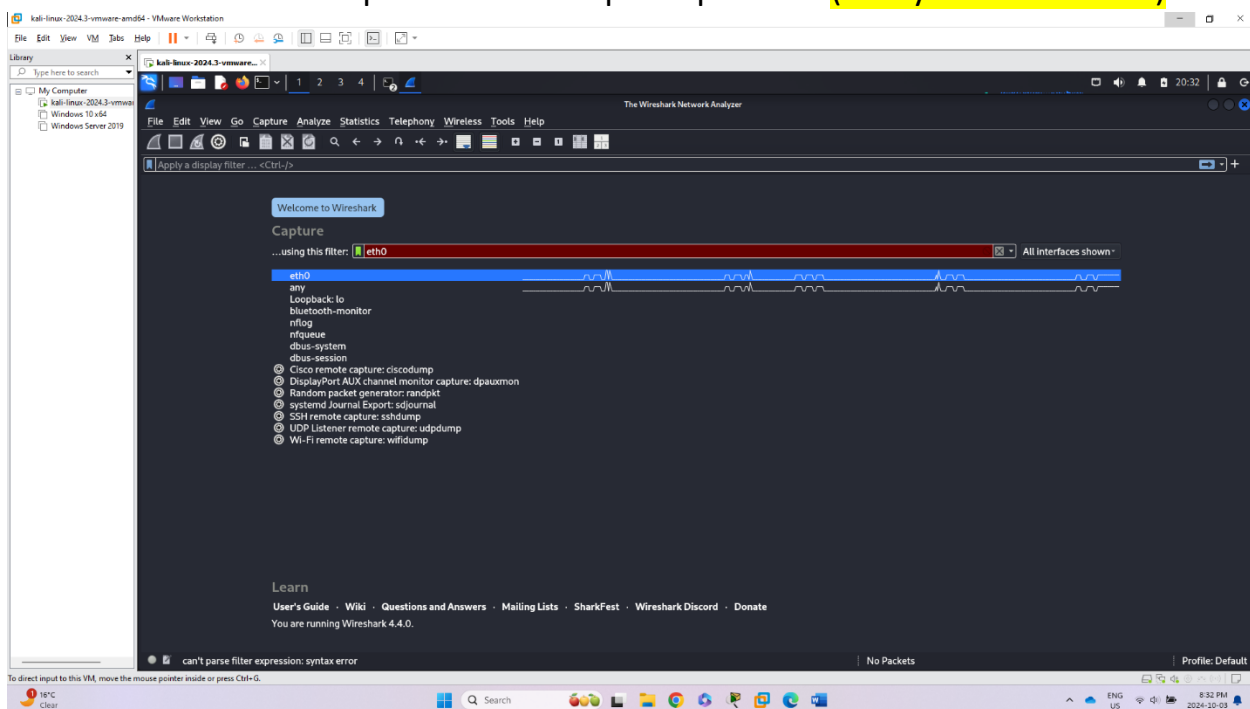
Documentation

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Starting Wireshark

1. Just as we did in class, start Wireshark from the menu or the command line to start capturing. Then, you need to choose an interface. If you are running the Wireshark on your laptop, you need to select WiFi interface. If you are at a desktop, you need to select the Ethernet interface being used. Note that there could be multiple interfaces. In general, you can select any interface but that does not mean that traffic will flow through that interface. The network interfaces (i.e., the physical connections) that your computer has to the network are shown.
2. Select an interface and press start to capture packets. (Add your screenshot)

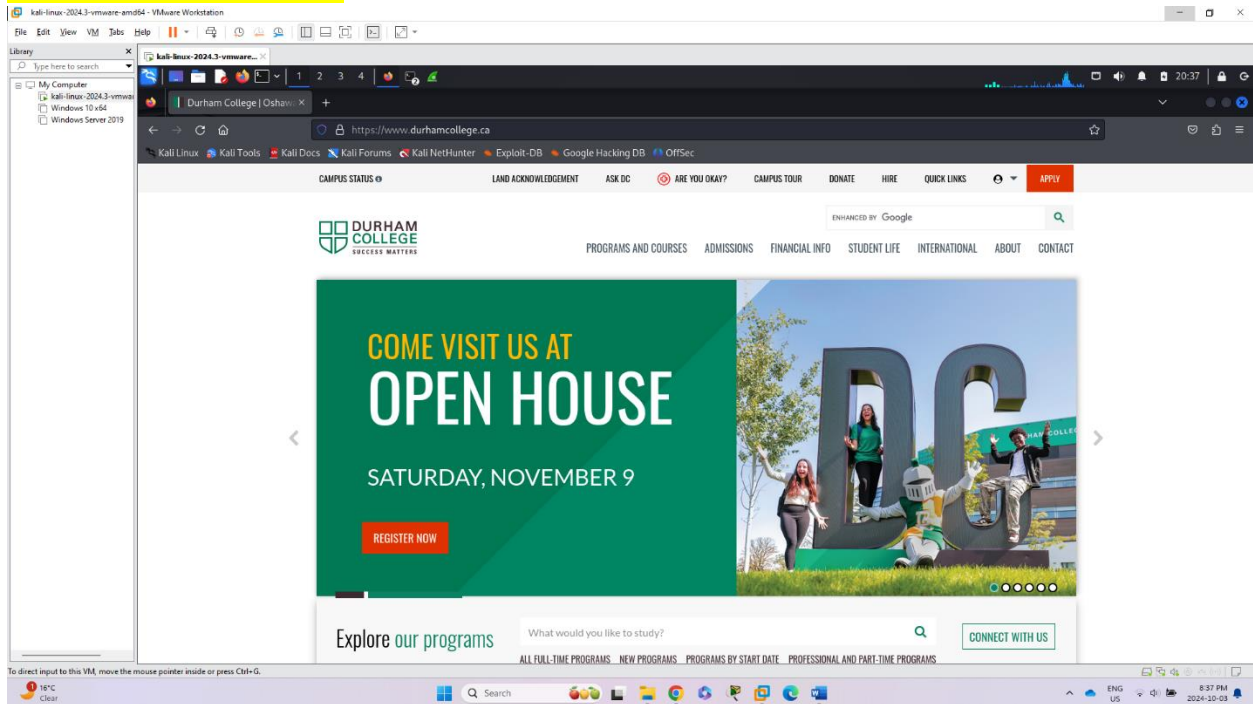


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3. In your browser, go to Durham Colleges website homepage by typing www.durhamcollege.ca. Add screenshot of the college's website visited.

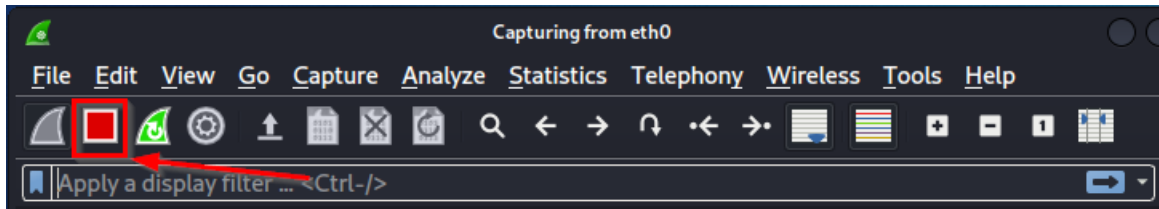
(Add your screenshot)



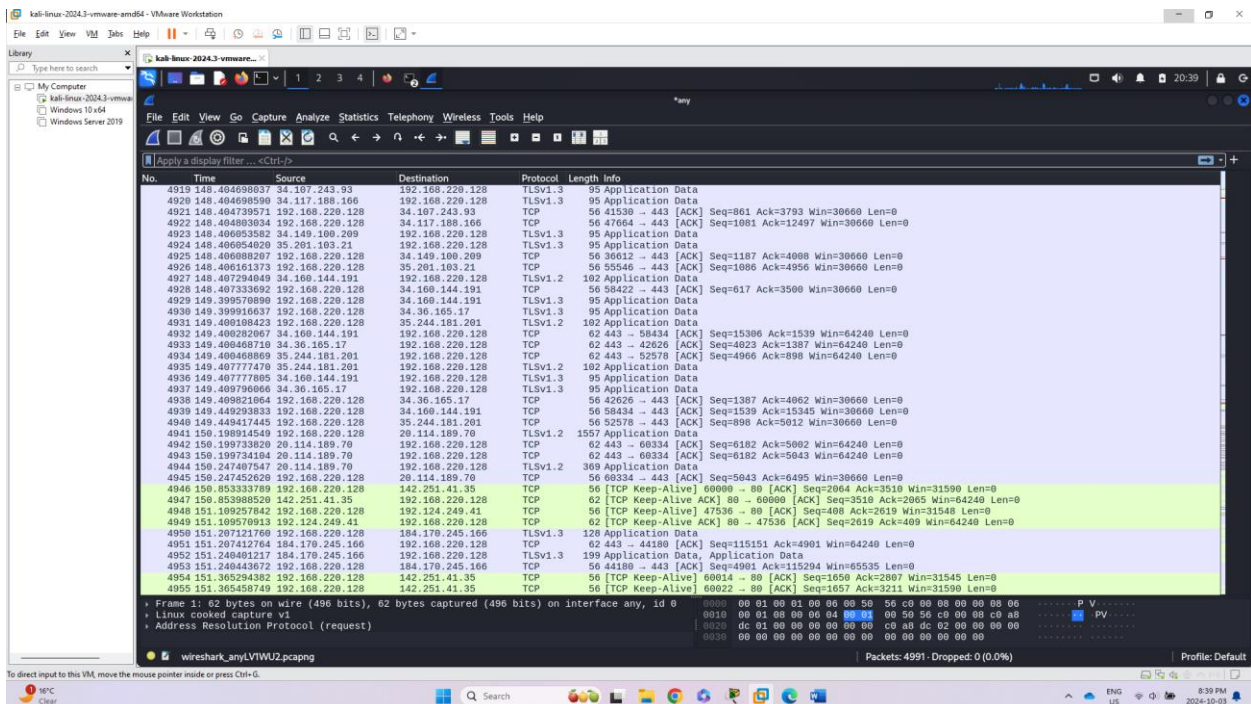
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- After your browser has displayed the <http://www.durhamcollege.ca> page, stop Wireshark packet capture by selecting stop in the Wireshark capture window. This will cause the Wireshark capture window to disappear and the main Wireshark window to display all packets captured since you began packet capture see image below:



(Add your screenshot)



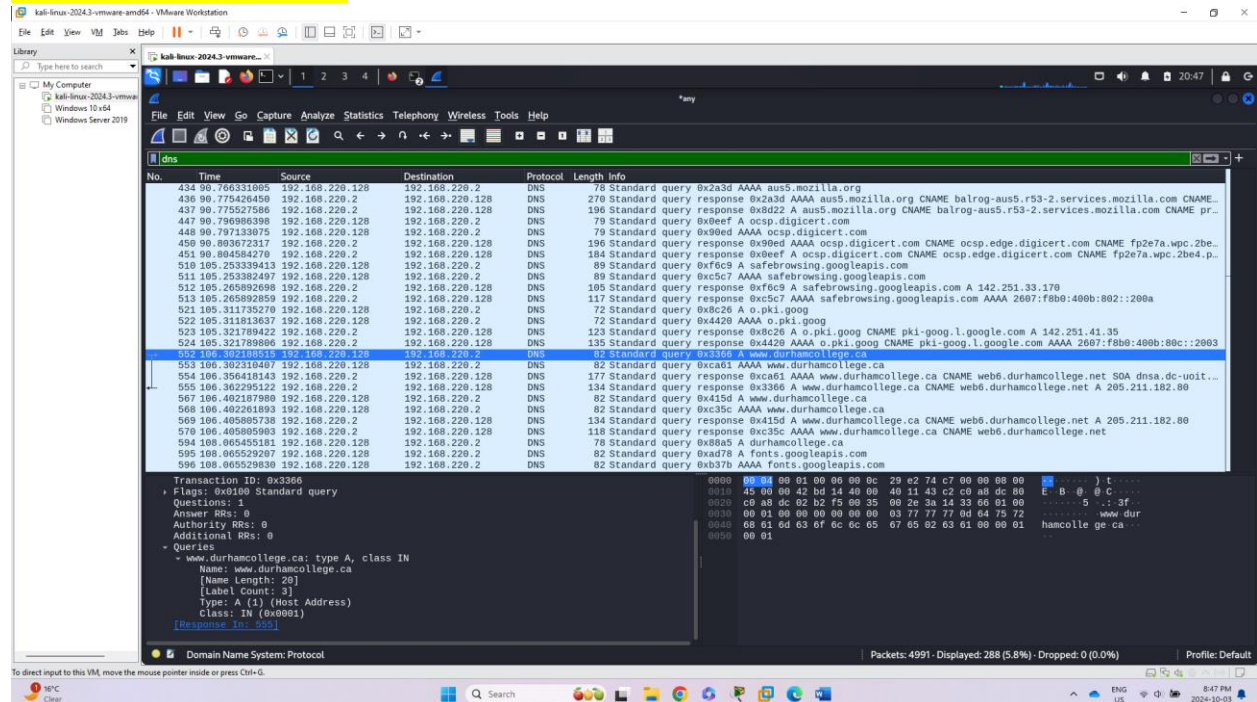
- Color Coding: You'll probably see packets highlighted in green, blue, and black. Wireshark uses colors to help you identify the types of traffic at a glance. By default, green is TCP traffic, dark blue is DNS traffic, light blue is UDP traffic, and black identifies TCP packets with problems — for example, they could have been delivered out-of-order.

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6. You now have live packet data that contains all protocol messages exchanged between your computer and other network entities! However, as you will notice the HTTP messages are not clearly shown because there are many other packets included in the packet capture. Even though the only action you took was to open your browser, there are many other programs in your computer that communicate via the network in the background. To filter the connections to the ones we want to focus on, we have to use the filtering functionality of Wireshark by typing “http” in the filtering field as shown below. Alternatively you can filter using dns to find the durhamcollege.ca website.

(Add your screenshot)



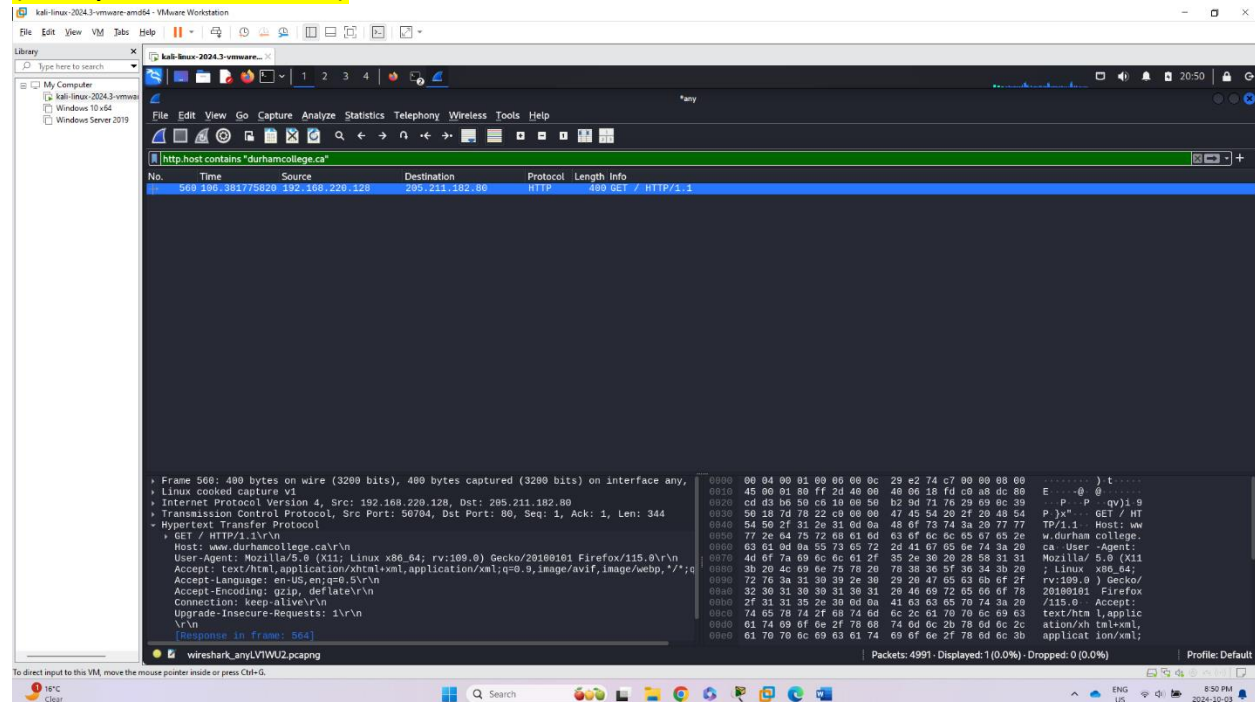
Notice that we now view only the packets that are of protocol HTTP. However, we also still do not have the exact communication we want to focus on because using HTTP as a filter is not descriptive enough to allow us to find our connection to <http://www.durhamcollege.ca>. We need to be more precise if we want to capture the correct set of packets.

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- To further filter packets in Wireshark, we need to use a more precise filter. By setting the filter (http contains durham), we are restricting the view to packets that have as an http host the www.durhamcollege.ca website.

(Add your screenshot)

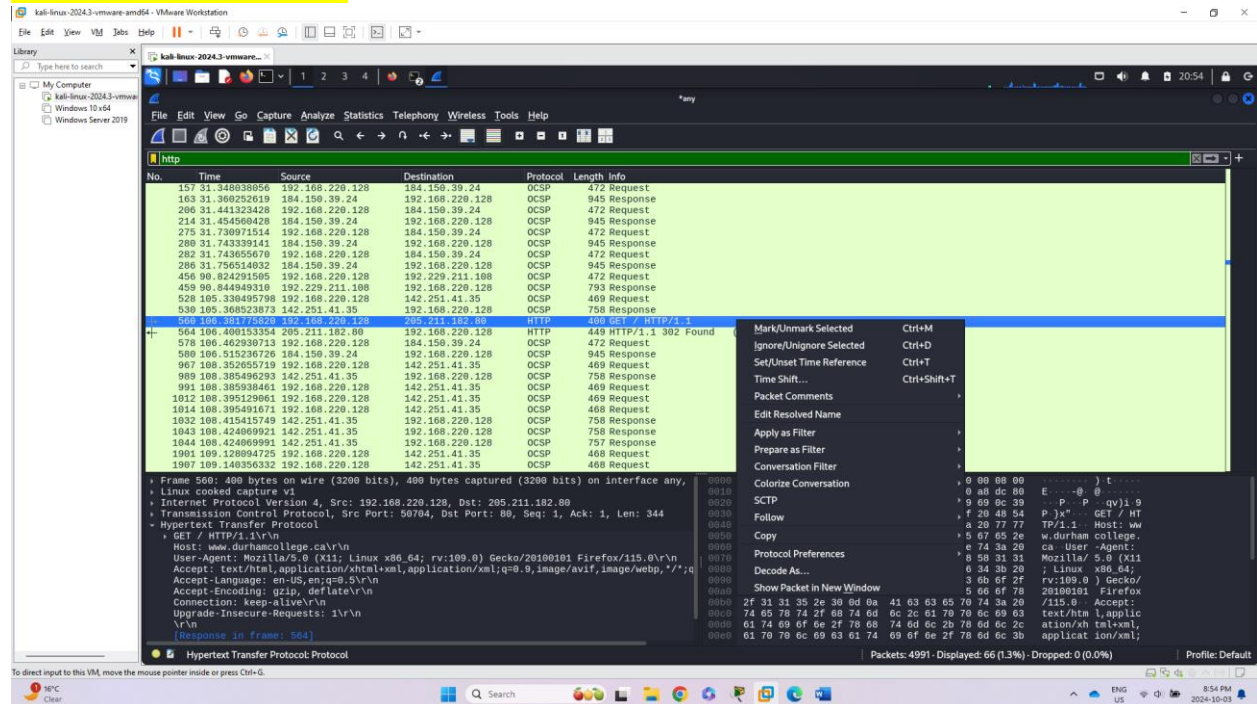


- Let's try now to find out what are those packets contain by following one of the conversations (also called network flows), select one of the packets and press the right mouse button (if you are on a Mac use the command button and click).

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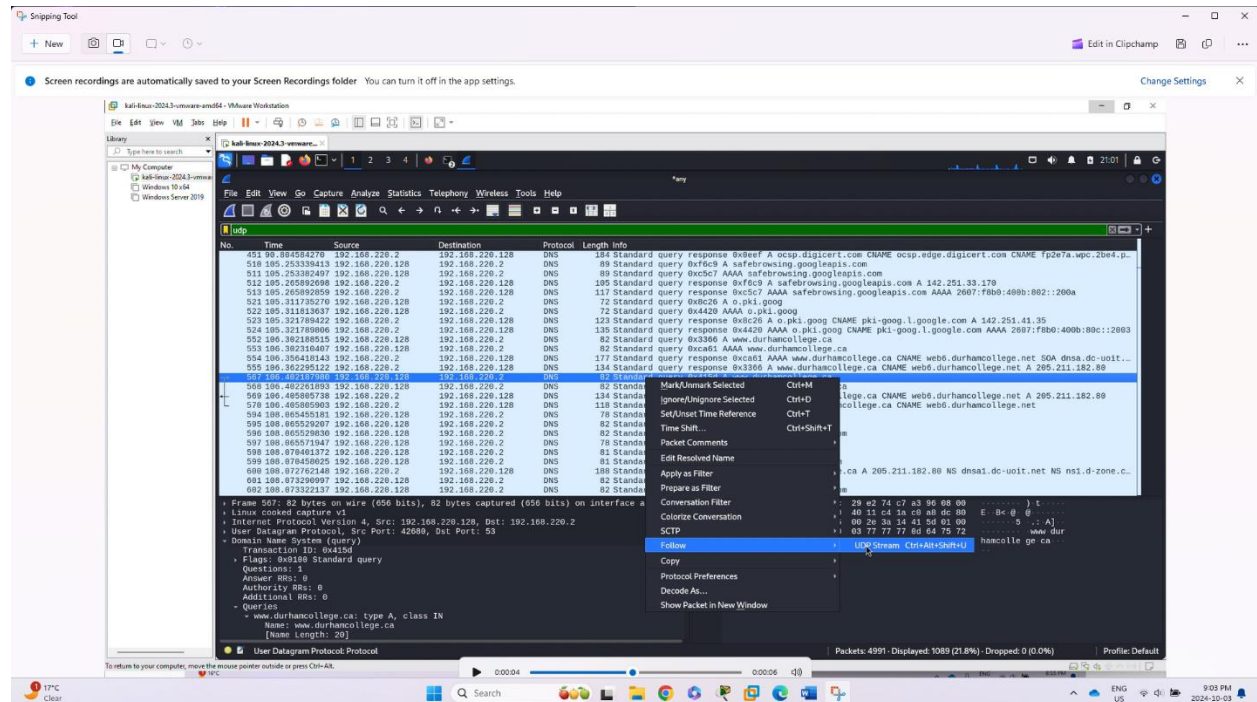
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(Add your screenshot)



Click on Follow UDP Stream

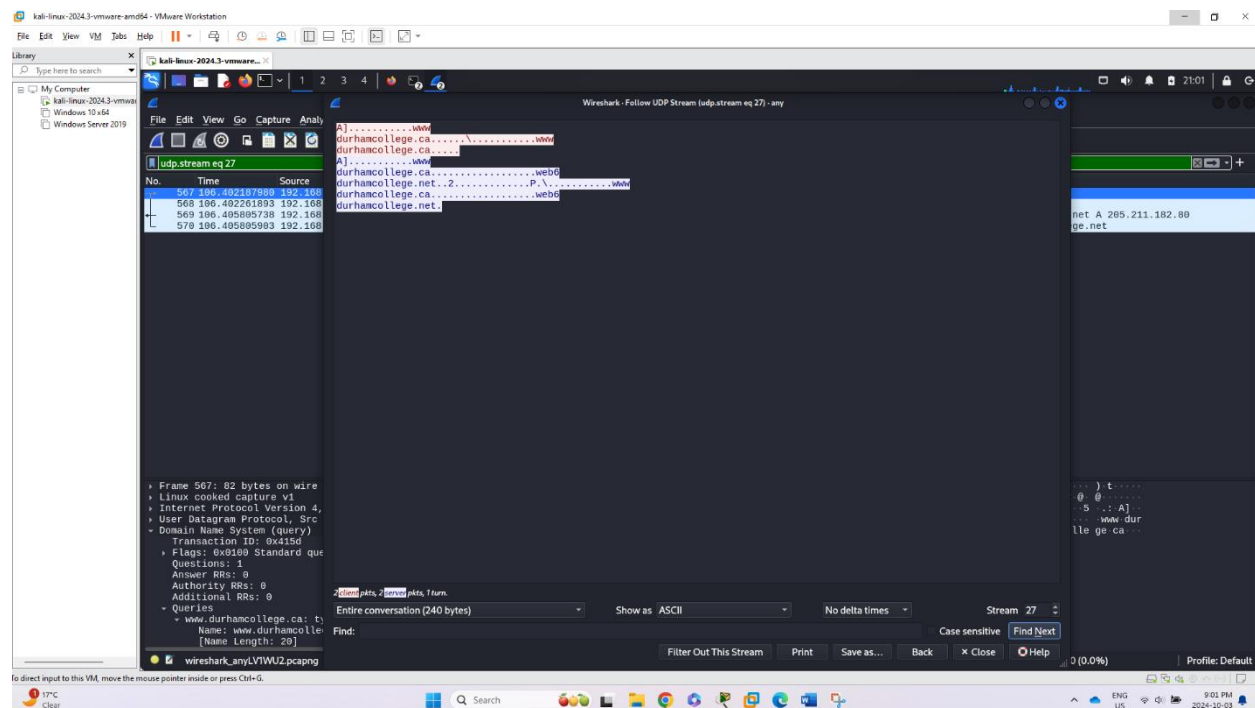
(Add your screenshot)



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(Add your screenshot)

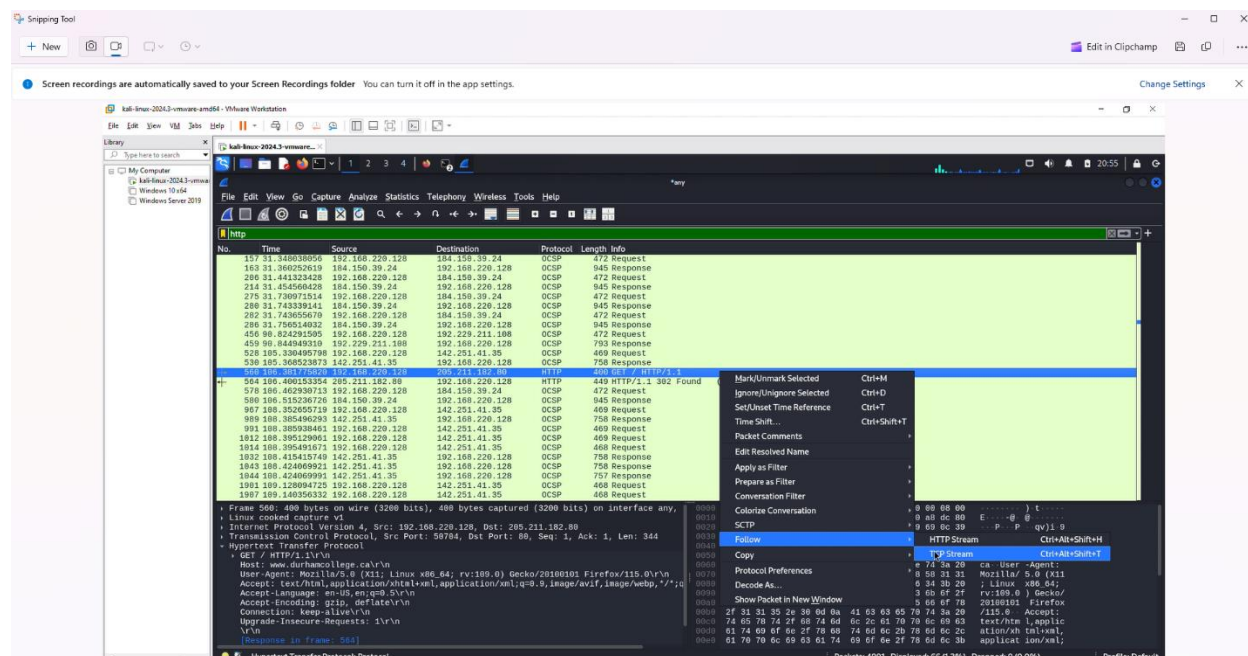


If we close this window and change the filter back to “http.host==www.durhamcollege.ca” and then follow a packet from the list of packets that match that filter, we should get something similar to the following screens. Note that we click on Follow TCP Stream this time.

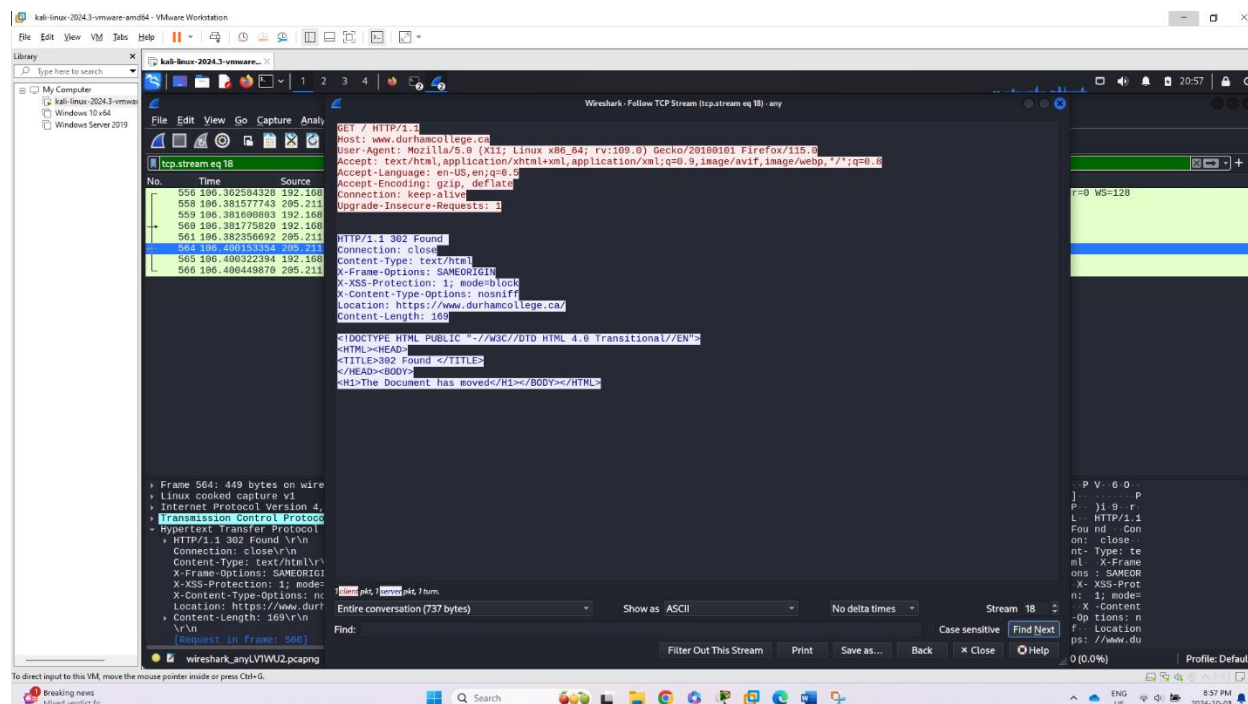
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(Add your screenshot)



(Add your screenshot)



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Things to Explore:

You are welcome to explore beyond the mandatory requirements if you wish.

General submission requirements

- Include an opening comment with your full name, date, and a short description.
- **Marks distribution:**
 - a) 10 Screenshots = 100 marks

Do not alter the sequence of steps of this document. Do not delete the words [Attach the screenshot] anywhere from the assignment. Keep the numbering same. Paste the screenshots wherever it's being asked- do not use screenshots of professors from lab manual. They are for your reference only and would be considered an Academic Integrity Breach. You should be not even submitting another student's screenshot- Academic Integrity alert.